Executive Report: Flavoured Water Demand Forecast Dashboard

AI in Operations: Problem And Exercise 2 · Nov 12, 2026 · Gagandeep Singh

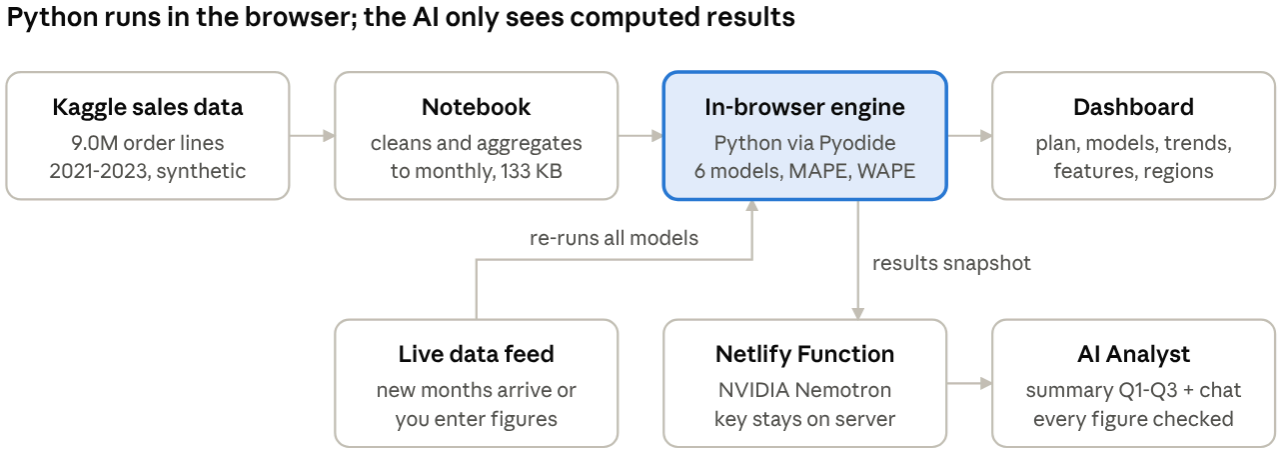
Bottom line

Plan **465,666 units** of the new flavoured water for January to March 2022 (160,396 · 144,874 · 160,396). The plan uses the Calendar-adjusted Naive model, which misses by **1.92% WAPE** in backtests against 2.37% for the Naive benchmark.

Demand after launch is stable: no significant trend (p = 0.212), and seasonality cannot be tested from one year. Days in the month explain 76% of month-to-month variation, so simple calendar-aware models win. Live dashboard: flavoured-water-forecast.netlify.app.

How it works

The dashboard forecasts monthly units of Volvic Touch (launched 2021) with six models: Naive, Seasonal Naive, Calendar-adjusted Naive, ARIMA, Random Forest and XGBoost. Users can change product, region, customer type, test window and horizon, release months one by one or enter their own figures. Every model re-runs in about a second and the AI Analyst re-interprets the new results.



how the dashboard works · data to engine to AI

Key results

The six models are close in the launch year. WAPE is from the rolling-origin backtest and ranks the models (lower is better); MAPE is on the test window.

Model	WAPE	MAPE	Role
Random Forest	1.78%	1.56%	Best on backtest
Calendar-adjusted Naive	1.92%	1.37%	**Planning model** (simplest within 0.5 points)
ARIMA	1.97%	1.38%	
Naive	2.37%	2.05%	Benchmark
XGBoost	2.83%	1.03%	Unstable across test windows
Seasonal Naive	not run	not run	Needs a full prior year

- **Out-of-sample check:** on the unseen months January to March 2022, Calendar-adjusted Naive was the most accurate model (2.65% WAPE vs 3.66% for Naive).
- **With three years of data,** ARIMA wins clearly (1.29% vs 3.42% WAPE for Naive).
- **Flat volume** (+0.21% a year, p = 0.538); revenue per day grows +2.77% a year (p < 0.001).

Recommendations

1. **Produce 465,666 units for January to March 2022,** preparing for the 95% range of 152,520 to 168,272 units in January. The model is almost unbiased (+0.09%).
2. **Plan per sales day:** February (28 days) needs about 10% less than a 31-day month.
3. **Allocate by region:** Saarland 6.7%, Hessen 6.5%, Hamburg 6.4%; demand is even across states.
4. **Keep simple models and re-forecast monthly;** re-test seasonality at 24 months.

Why the numbers can be trusted

- **Fair evaluation:** one chronological split and rolling-origin backtest for all models.
- **One engine, tested:** notebook and dashboard share the same Python code; 45 Python, 70 dashboard and 20 browser tests pass.
- **Grounded AI:** the AI sees only computed results; answers quoting numbers not in the data are rejected, with a rule-based fallback.
- **Limits:** synthetic data with no real seasonality; only 12 launch-year data points.

Try it: open the live dashboard, then **Overview**, **Live Data Feed** and **AI Analyst › Chat**.

Code, notebook and test report are in the GitHub repository (public; open access for anyone).